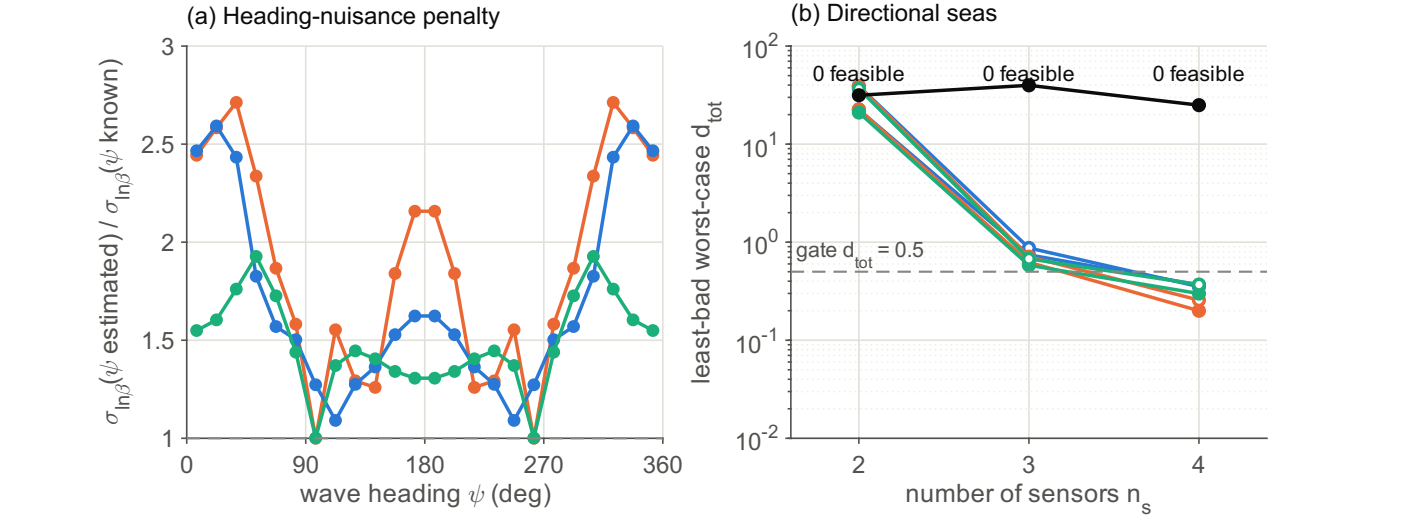

(a) Heading-nuisance penalty
$\sigma_{\ln\beta}(\psi$ estimated) / $\sigma_{\ln\beta}(\psi$ known)
wave heading ψ (deg)
(b) Directional seas
least-bad worst-case d_{tot}
number of sensors n_s
0 feasible
0 feasible
0 feasible
gate d_{tot} = 0.5
D1 optimum, n_s = 2
D3a robust, n_s = 2
Reference configuration
ϕ = 30°, f_2 = 25%
ϕ = 60°, f_2 = 25%
ϕ = 90°, f_2 = 25%
ϕ = 30°, f_2 = 10%
ϕ = 60°, f_2 = 10%
ϕ = 90°, f_2 = 10%
$\cos^{2s}$ spread, s = 10